Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B.Tech.- Aerospace Engineering

Semester: 3^{rd}

Name of the Paper: Aero-Fluid Mechanics

Paper Code: TAS-304

Time: 3 Hours

Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1		(20marks)
(a)	Define the following terms: a) Buoyancy b) Metacentric height c) Newton's Law of Viscosity. d) Surface Tension.	CO1
(b)	Which law states that "The intensity of pressure at a point in a fluid is same in all direction"? Derive the law with proper figure.	CO2
(c)	The space between two square flat parallel plates is filled with oil. Each side of the plate is 720 mm. The thickness of the oil film is 15 mm. The upper plate, which moves at 3 m/s requires a force of 120 N to maintain the speed. Determine (i) The dynamic viscosity of the oil (ii) The kinematic viscosity of oil if the specific gravity of oil is 0.95	CO2
Q2		(20 marks)
(a)	Define the following terms: i) Eulerian and Langragian approach of fluid flow ii) Stream Function and Velocity Potential Function	CO2
(b)	Prove that vorticity at a point in a flowing fluid is the curl of the velocity vector.	CO2
(c)	The velocity components in a two dimensional velocity field for an incompressible flow are, $u=(y^3/3) + 2x - x^2y$ $v= xy^2 - 2y - (x^3/3)$ Show that these functions represent a possible case of irrotational flow.	CO3
Q3		(20 marks)
(a)	Explain the principle behind continuity equation and derive the integral form of continuity equation.	CO3
(b)	Derive Bernoulli's equation.	CO3
(c)	Water flows in a circular pipe. At one section the diameter is 0.3 m the static pressure is 260 kPa gauge, the velocity is 3 m/s and the elevation is 10 m above ground level. The elevation at a section downstream is 0 m and the pipe diameter is 0.15 m. Find the gauge pressure at the downstream section. Frictional effect may be neglected. Assume density of water to be 999 kg/m^3.	CO3

Q4		(20 marks)
(a)	Explain the steps involved in solving a problem using Buckingham pi theorem.	CO4
(b)	With the help of a neat sketch, explain Reynolds Experiment.	CO4
(c)	The time period 't' of a pendulum depends upon the length of the pendulum 'L' and acceleration due to gravity 'g'. Derive an expression for the time period.	CO4
Q5		(20 marks)
(a)	Explain flow separation with neat sketches. Explain various methods used for preventing flow separation.	CO5, CO6
(b)	Discuss boundary layer development on a flat plate.	CO 5, CO6
(c)	Find the ratio of displacement thickness and momentum thickness to energy thickness for the velocity distribution in boundary layer given by $\frac{u}{U} = 2\frac{y}{\delta} - (\frac{y}{\delta})^2$	CO6